Multi-Feature Secure ATM System using Java

Problem Statement : To develop a secure ATM system in Java that allows users to access their bank account using a PIN. The program should verify the user's PIN with a limited number of attempts and provide ATM functionalities such as checking balance, withdrawing money, and transferring money while validating user input and preventing invalid transactions.

Objectives

- To understand the use of **variables and appropriate data types** in Java.
- To implement **conditional statements (if–else and nested conditions)**.
- To use **loops (while and do-while)** for repeating operations such as menu display and PIN attempts.
- To practice **arrays and objects** for storing multiple user accounts.
- To handle **user input using Scanner**.
- To develop a **menu-driven ATM program** that performs banking operations securely.

SOURCE CODE:

```
import java.util.Scanner;

class User {
String name;    int
pin;    double
balance;

   User(String name, int pin, double balance) {
this.name = name;       this.pin = pin;
      this.balance = balance;}}

public class ATM {

   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);

      User[] users = {              new
User("Usman", 1111, 1000),              new
User("Ali", 2222, 900),             new
```

```
User("Eman", 3333, 800),                new
User("Manahil", 4444, 700),
            new User("Mehran", 5555, 600) };

      int attempts = 0;
User currentUser = null;

      while (attempts <= 3) {

         System.out.println("Enter Pin:");
int enteredPin = sc.nextInt();
for (User user : users) {                if
(user.pin == enteredPin) {
currentUser = user;
                break;} }

         if (currentUser != null) {
            System.out.println("Login Successful!");
break;}           else {                attempts++;
            System.out.println("Wrong Pin. Attempts Left: " + (3 - attempts));}  }

      if (currentUser != null) {

         int choice;

         do {

            System.out.println("\nATM Menu");
            System.out.println("1. Check Balance");
            System.out.println("2. Withdraw Money");
            System.out.println("3. Send Money");
            System.out.println("4. Exit");

            System.out.println("Enter Choice:");
            choice = sc.nextInt();

            if (choice == 1) {

               System.out.println("Your balance: " + currentUser.balance);

            } else if (choice == 2) {
```

```
                System.out.println("Enter amount:");
double amount = sc.nextDouble();

                if (amount <= currentUser.balance) {
currentUser.balance -= amount;
                    System.out.println("Remaining balance: " + currentUser.balance);
                } else {
                    System.out.println("Insufficient balance");
                }

            } else if (choice == 3) {

                System.out.println("Enter amount to send:");
double amount = sc.nextDouble();

                if (amount <= currentUser.balance) {

                    currentUser.balance -= amount;
System.out.println("Money Sent");}

                 else {
                    System.out.println("Insufficient balance");} }

        } while (choice != 4);

      } else {
        System.out.println("Login Failed");} sc.close();}}
```

Output:

```
Enter Pin:
1234
Wrong Pin. Attempts Left: 2
Enter Pin:
1111
Login Successful!

ATM Menu
1. Check Balance
2. Withdraw Money
3. Send Money
4. Exit
Enter Choice:
1
Your balance: 1000.0

ATM Menu
1. Check Balance
2. Withdraw Money
3. Send Money
4. Exit
Enter Choice:
2
Enter amount:
690
Remaining balance: 310.0

ATM Menu
```

```
ATM Menu
1. Check Balance
2. Withdraw Money
3. Send Money
4. Exit
Enter Choice:
3
Enter amount to send:
100
Money Sent

ATM Menu
1. Check Balance
2. Withdraw Money
3. Send Money
4. Exit
Enter Choice:
4
```